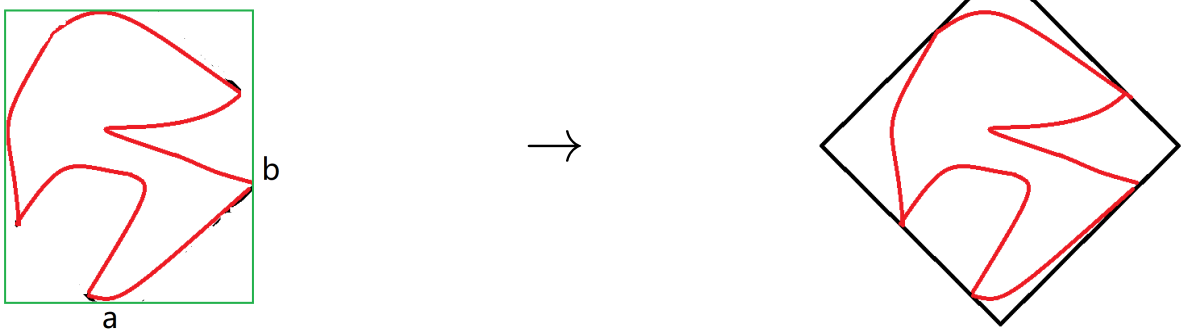
A: The Intermediate Value Theorem

The proof: To see this you first draw a rectangle satisfying the given property. Then you rotate the rectangle continuously, to get new rectangles (with different side lengths a and b satisfying the same requirement). As you rotate, the value of $a - b$ changes continuously. Moreover, it will change sign after rotating 90 degrees. So it must attain the value 0 during the rotation.



The square peg problem

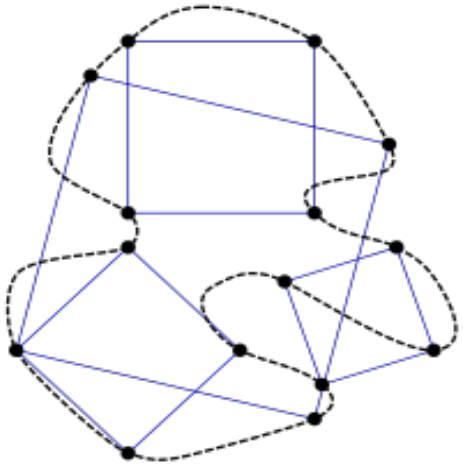
Before we move to next topic, we mention a similar problem in planar geometry. It is proposed by O. Toeplitz in 1911, and is known as the *inscribed square problem*, or the *square peg problem*. It is still unsolved today!

Conjecture (Toeplitz)

On every continuous simple closed planar curve, one can find four points that are the vertices of a square.

The square peg problem

Here is an example on which you can find four such squares:



The square peg problem

Some developments on the square peg problem:

1. Emch (1916): true for smooth convex curves, piecewise analytic curves (by using some kind of IVT argument)
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The square peg problem

Some developments on the square peg problem:

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3. Vaughan 1977: On every continuous simple closed planar curve, one can find four points that are the vertices of a rectangle.
    1. Idea: use the topology of the Möbius strip!
4. Greene-Lobb 2020: For any rectangle, on any smooth simple closed planar curve, one can find four points that are the vertices of a rectangle.
    1. Idea: use symplectic topology!

B: The Extremal Value Theorem

For the second topic, again we start with a baby theorem:

Baby-Theorem

Every finite subset of $\mathbb{R}$ contains its supremum/infimum.

B: The Extremal Value Theorem

In mathematical analysis, we learned

Theorem (The Extremal Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f attains both a maximal value and a minimal value.

B: The Extremal Value Theorem

In mathematical analysis, we learned

Theorem (The Extremal Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f attains both a maximal value and a minimal value.

- Easy to see
  - The theorem fails if you replace $[a, b]$ by (a, b) or $[a, +\infty)$.
  - The theorem holds if you replace $[a, b]$ by $[a, b] \cup [c, d]$.

B: The Extremal Value Theorem

- Topological property behind the Extremal Value Theorem:

$$\boxed{\textit{compactness} \text{ of } [a, b].}$$

(We will see: compactness is a generalization of finiteness.)

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- We can easily extend the EVT to

Theorem (The Generalized Extremal Value Theorem)

Let X, Y be topological spaces, and let $f : X \to Y$ be a continuous map. Then for any compact subset $A \subset X$, the image $f(A)$ is a compact subset in Y.

Compactness

Compactness is one of the most important conception in topology. It is widely used in many other branches of mathematics, especially in analysis.

Dirichlet't principle

Here is some history: In 1856-1857, when lectured on potential theory, Peter Dirichlet (1805-1859) proposed a method, known as *Dirichlet't principle*[8], to solve the equation

$$\Delta u(x, y, z) = 0$$

on a region $\Omega \subset \mathbb{R}^3$ with prescribed boundary value f on the boundary surface $S = \partial\Omega$.



[8] named by Riemann, but the method was used earlier by Green, Gauss, ...

Dirichlet's principle

- To solve this he considered the integral

$$U = \int_\Omega \left[(\frac{\partial u}{\partial x})^2 + (\frac{\partial u}{\partial y})^2 + (\frac{\partial u}{\partial z})^2 \right] dV.$$

which is always non-negative. Dirichlet concluded that there must be at least one function u on Ω for which the integral reaches a minimum. Then by using a standard variational argument one can show that the minimizer satisfies $\Delta u = 0$.

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- Wait! As criticised by Weierstrass, there was a problem: The existence of a greater lower bound for the values of the integral does not necessarily imply the existence of a minimizer. To guarantee this, one need compactness (where we view functions as points).

Compactness: Arzela-Ascoli

The classical Arzela-Ascoli theorem is one of the first attempts trying to solve this type of problems. They justified Dirichlet's principle under extra assumptions:

Theorem (Arzela-Ascoli)

Let $\{f_n\}$ be a uniformly bounded and equicontinuous sequence of real-valued continuous functions defined on $[a, b]$. Then there exists a subsequence that converges uniformly.

Compactness: Arzela-Ascoli

In this course we will prove a much general version:

Theorem (Generalized Arzela-Ascoli Theorem)

Let $\mathcal{F}$ be a point-wise bounded and equicontinous family of functions defined on a compact space X. Then for any sequence in $\mathcal{F}$, there exists a subsequence that converges uniformly.

Compactness: The Blaschke selection theorem

• Arzela-Ascoli theorem is widely used in functional analysis and partial differential equation to prove the existence of a limit. Here we mention an equivalent form of the Arzela-Ascoli theorem in convex geometry:

Theorem (The Blaschke selection theorem)

Given a sequence $\{K_n\}$ of convex sets contained in a bounded set, there exists a subsequence $\{K_{n_m}\}$ and a convex set K such that K_{n_m} converges (in a sense that will be clear later) to K.

Compactness: The Blaschke selection theorem

• The Blaschke selection theorem can be used to prove the existence of a solution to many geometric problems (but the solution is still unknown). E.g.:

- Lebesgue's universal covering problem: the convex shape of smallest area that can cover any planar set of diameter one. (Note that a circle with diameter 1 is NOT a solution because it can't cover the Reuleaux triangle.) (Known: it should has area between 0.832 and 0.8440935944)

C: Separation properties

- Again we start with a baby theorem:

Baby-Theorem

Suppose $a < b < c < d$. Then one can find a continuous function $f : \mathbb{R} \to \mathbb{R}$ s.t. $f([a, b]) = 0, f([c, d]) = 1$.